Assesment - 1

Section A : Concept Application

1. Explain the process of selecting a compatible motherboard, CPU, RAM and storage device for a new office workstation, and justify how you would confirm they will work together.

→

* CPU & motherboard :-
  match the physical socket like intel LGA 1700 or AMD AM5 and ensure the chipset supports the processor generation.

* RAM :-
  Ensure the memory type (like DDR4 or DDR5 and slot form factor (UDIMM) match both the motherboard & CPU space.

* Storage Device :-
  verify that the motherboard has m.2 slot supporting the drive's protocol (NVme PCIe vs SATA

* To confirm :-
  check the motherboard's manufacturer's offical CPU support list and memory qualified vendor List (QVL) or run your exact parts through pcpartpicker.

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2. Explain the difference between HDD, SSD and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

→

★ HDD :-

Uses spinning magnetic platters (~150 MB/s)
Extremely slow, cheap storage.

★ SATA SSD :-

Flash memory using SATA cable (~550 MB/s)
moderate speed, silent.

★ NVme SSD :-

Flash memory Attached directly to PCIe lanes (3500- 7500+ MB/s).

★ Recommendation :- NVme M.2 SSD

Higher Input/output operations per second (IOPS) and low latency eliminate bottlenecks during multi-tasking, providing near - instant OS boot, rapid file transfers and minimal application load times.

9. Explain the role of GPU and the cooling system in maintaining performance and Stability under this kind of sustained load.

→ * GPU :

It Handles heavy visual rendering, complex 3D math and parallel data tasks.

- offloads massive visual workloads from the CPU so the system runs smoothly without lagging or crashing under heavy stress.

* cooling system :-

Pulls heat away from the CPU and GPU using fans, heatsinks or liquid loops.

- Prevents where components intentionally slow down to avoid melting (thermal throttling) and stops unexpected thermal shutdowns during long, Intense sessions.

4. Explain why hardware compatibility matters when assembling a computer and identify potential issues that could arise from incompatible components.

→ Computers need all parts to communicate, fit physically and share power safety. If components don't match, the system simply won't turn on or can be permanently damaged.

* Potential issues from incompatible components :-

* Physical fit failures :-
The CPU won't fit into the wrong motherboard socket, or the graphics card is too long for PC case.

* mismatch RAM/storage :-
plugging DDR5 or DDR4 into another slots that's physically fails, and buying an NVme SSD for a board without m.2 slots leaves you with unusable storage.

* power & overheating :-
A weak power supply unit (PSU) causes random shutdowns during gaming, while an undersized CPU cooler triggers thermal throttling & system crashes.

* System crashing :-
mismatched parts lead to Blue screens of Death (BSOD), boot loops, or a completely dead PC that won't show a display.

5. Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

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* choose an appropriately rated PSU :-

* wattage :-
calculate your CPU and GPU power needs, then add 30% for a safe buffer.

* Efficiency :-
pick an 80 plus Bronze or Gold certified unit from a reputable brand.

* checks Before powering on :-

1. click All cables :- push 24-pin main power, CPU, GPU and SATA cables in until they fully click.

2. clear the Fans & Tuck away loose wires so they don't block fan blades.

3. Flip the switch :- Flip the switch on the back of the PSU to "I" (ON).

6. Explain the checks you would perform to confirm the system passes POST successfully, and what you would do it if did not.

→

* Confirming successful POST :

* Visual output : The display turn on and shows the BIOS splash screen.

* Diagnostic LEDs : motherboard debug LEDs (CPU, DRAM, VGA, BOOT) cycle through and full all turned

* Peripherals :- Keyboard lights illuminate and fan speeds drop from full speed to quiet idle.

* IF POST Fails :

1. Read Debug LEDs : Identify which status LED stays lit (CPU, DRAM or VGA).

2. Reseat RAM : Re-insert memory sticks firmly into recommended slots until latches click.

3. Check Power cables : verify 24-pin ATX & 8-pin CPU connectors are fully seated.

4. Clear CMOS : Remove motherboard coin battery for 5min to reset BIOS settings.

5. Minimal Boot : Disconnect non essential hardware, running only 1 RAM stick, CPU + PSU.

Section - B

Practical coding Tasks

1. Component Identification & Functions :

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Components	Primary Functions	Key Specification
Motherboard	connects all hardware & routes power & data.	socket & chipset For ex :- LGA 1700, AM5
CPU	Executes instructions & processes system logic	cores, threads & clock speed (GHz)
RAM	provides fast temporary memory for active apps.	capacity & type For ex : 16 GB DDR5.
GPU	Renders visual outputs & graphics tasks	model or iGPU type & VRAM
Storage	Retains OS, apps & files permanently.	Interface & capacity For ex : (1TB NVMe m.2)
Cooling	Dissipates CPU heat to prevent throttling	Type (Air or AIO liquid) & Fan size.

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2. Motherboard ports & socket identification :

→ Internal sockets & slots :

CPU socket : Houses the processor (e.g. LGA 1700, AM5)
DIMM slots : Holds RAM modules (DDR4/DDR5)
PCIe x16 slot : connects primary expansion card (GPU)
PCIe x1/x4 slot : connects smaller add-on cards (Wi-Fi, Sound)
M.2 slots : secures fast NVMe SSD storage.

* Internal power & Headers :

- 24-pin ATX : main motherboard power point.
- 8-pin EPS : Dedicated CPU power input.
- SATA ports : connects standard 2.5/3.5 drives.
- Front panel Header : connects case power button, reset switch & LEDs.

* Rear I/O ports :

- HDMI / Display ports : video output for monitors.
- USB ports (Type-A / Type-C) : connects peripheral & external storage.
- Ethernet : wired network connection.
- 3.5mm Audio Jacks : microphone, headphone & speaker connections.

3. RAM & Storage Installation :-

→ ESD Safety :-
Unplug power cable, hold computer's power button for 5s to drain residual power & touch bare case metal to ground yourself.

* RAM :-

Remove :- push down slot side to eject module.

Reinstall :- Align bottom notch with slot key, press lightly down until latches click shut.

* Storage :-

NVme M.2 :- Remove mounting screw, pull drive out at 30° angle. Reinsert at 30°, press flat, tighten screw.

SATA :- Unplug power / data cables. Reinstall by plugging both cables back in flush.

* Confirmation :-
Boot into BIOS (Del+F2) and verify RAM capacity & storage drive model display under System summary.

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4. Full Desktop PC Assembly:

→ 1. Preparation & Safety: workspace & Anti-static.
- Set up on clean, non-conductive surface.
- Ground yourself by touching bare metal or wearing an anti-static wrist strap.

2. Out of case Installation: core components installed on motherboard first.
- CPU: Unlatch socket, align golden corner triangle lower gently and lock lever.
- RAM: Align stick notch with slot key press firmly down until both side latches click shut.
- M.2 SSD: Insert into at 30° angle, Press flat & secure with mounting screw.
- CPU cooler: Apply thermal paste, secure cooler in a cross pattern, and connect fan to CPU-Fan.

3. Case mounting: secure hardware into chassis.
- PSU: Install power supply into bottom chamber & secure with 4 screws.

- Motherboard: verify standsoffs align, place board, and tighten mounting screws.

4. Cabling & GPU Installation: power & Expansion

- connect 24-pin ATX, 8 pin CPU and front panel cables securely.
- GPU: Remove case slot covers, Press GPU into top PCIex16 slot until latches, screw into frame and connect PCIe power cables.

5. Power on & POST verification & Initial Boot

- plug display cable into GPU, connect power and press case power button.
- verify fans spin, motherboard debug LEDs cycle off, and BIOS splash screen displays.

Section - C

mini capstone project

mini project : workstation deployment & Boot - Failure Diagnosis.

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1. Top 3 Boot - Failure Diagnostics :-

- RAM :-
 Unseated modules block post, lighting DRAM debug LEDs or causing continuous reboot loops.

- Power cables :-
 Loose 24-pin ATX or 8-pin CPU cables prevent motherboard ignition.

- CPU & Socket :-
 Bend pins, improper seating or missing CPU-FAN connections make instant safety shutdowns.

2. motherboard - CPU compatibility :

- Socket match : physically pin must match
For ex : intel LGA 1700 vs AMD Am5)

- BIOS Firmware : verify matching sockets still require updated motherboard BIOS microcode to recognise newer CPU gen

3. RAM & Storage verification :

- Physical check : confirm RAM clips clicked shut and m.2 SSD are screw down flat.

- BIOS confirmation : verify total memory capacity and drive model names show in the BIOS summary.

4. Deployment checklist :

- Check stand-offs & socket Alignment :
Ensure motherboard stand-offs match the board holes to prevent short circuits. Double check CPU alignment before latching to avoid bent pins.

- Verify power & Thermal contact :
confirm thermal paste is applied, the cooler cover is removed, and all power cables (24 pin, 8 pin) are firmly clicked into place.

- Run out-of case POST Test :-
Test CPU, RAM and GPU on the motherboard box before mounting inside the case to catch hardware DOA (Dead on Arrival) issues early.

5. SSD vs HDD Justification :-

- Zero Bottlenecks :-
SSDs load files and programs instantly, keeping lab software running smooth with zero lag or freezing.

- No moving parts :-
No mechanical spinning disks means lower risk of crash or data loss from table bumps and lab vibrations.

- Higher productivity :-
Near instant boot times and quick data transfers keep lab workflows fast without time wasted waiting on loading screens.

Section D

AI- Augmented Learning

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Step1 :- AI Generated Diagnostic checklist :

→ Step:1 :- PSU & Electrical check :-
- AC cord plugged in & PSU rocker switch set on
- Internal 24-pin ATX and 8-pin CPU cables Fully latched.

Step 2 :- motherboard Diagnostics :-
- check Debug LEDS (CPU, DRAM, VGA)
- verify case front panel power sw cable connect

Step 3 :- memory Diagnostics :-
- Reseat Ram in primary slots until latches click
- Test boot with a single RAM stick in slot A2.

Step 4 :- Storage Diagnostic :-
- confirm m.2 SSD is seated Flat and screwed down.
- Enter BIOS (Del + F2) to verify drive detection

ABC Solution Application step 3 :- catched unseated RAM or improper slot placement the primary cause of Fans spinning with no display on newly assembled lab PCs.

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Step 2: Manual corrections (safety & logic fixes):-

- Missing ESD / power drain precaution (safety flaw):-

Step 3 instructed opening the case and reseating RAM without safety protocols.

* Fix :- Insert mandatory step :-

Unplug AC power, hold case power button 5s to drain capacitors and ground off yourself on metal chassis before touching internal components.

- Out of order front panel check (logic flaw):-

Step 2 placed power SW checks after debug LED's but a loose power switch prevents the PC from attempting to turn on at all.

* Fix :- move front panel power SW verification directly to step 1 alongside PSU checks.